ABHISHEK MAHESHWARAPPA

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A passionate Data Scientist and data science blogger on medium with a liking to solve Business problems, developing scalable production-ready models for various use cases. Experience in working with Computer Vision, Natural Language Processing and Classical machine learning algorithms and worked on the ML-OPS side in deploying machine learning models into production.

WORK EXPERIENCE

Data Science Manager Ahold Delhaize, USA Jan 2024 – Present

- **Enterprise Data Science Strategy & Roadmap:** Defined and executed the AI transformation roadmap, aligning data science initiatives with business objectives to drive innovation and operational efficiency.
- **Strategic Planning Leadership:** Led annual strategy discussions with product teams to prioritize and plan the next year's initiatives for the Data Science organization.
- **Graph Machine Learning Innovation:** Spearheaded a new initiative to adapt graph-based machine learning models for enhancing personalization and cross-sell opportunities in grocery e-commerce.
- **Promotions & Loyalty Personalization:** Led the development of personalized coupon recommendations, resulting in an increase in coupon clip rate from 30% to 34%.
- **Search Optimization:** Upgraded semantic search systems into a hybrid search engine with personalization layers, driving search Add-to-Cart rates from 60% to 63%.
- **Executive Presentations:** Presented key Data Science initiatives and outcomes during Quarterly Business Reviews (QBRs) to the CIO and executive leadership.
- **OKR Development:** Developed and cascaded OKRs (Objectives and Key Results) for the Data Science team by aligning them directly with enterprise-level OKRs.
- **Model Impact Measurement:** Initiated and led efforts to define model-specific KPIs and collaborated with the Analytics & Reporting team to measure the business impact of deployed machine learning models.
- **Chatbot Development:** Built HR chat assistants using Databricks LLaMA, Retrieval-Augmented Generation (RAG), embedding models, and prompt engineering; delivered interactive Streamlit interface that boosted HR productivity by 30%.
- **Cross-Functional Collaboration:** Partnered with product managers (Offer Delivery, Personalization, Browsing, Shoppable Content) and stakeholders to capture requirements, prioritize DS backlog, and co-define the team roadmap.
- **Architecture & Solution Design:** Led end-to-end solution architecture, code reviews, and project planning for data science projects, ensuring scalable, maintainable pipelines in Databricks and Azure.
- **Evaluation Toolkit:** Designed and managed a modular evaluation framework for recommendation algorithms and semantic search, cutting model evaluation time by 8 hours per project.
- **Recommendation Systems:** Developed a neural collaborative filtering-based personalization service, increasing add-to-cart actions by 35%, and built a recipe recommendation engine that lifted average order value by 20%.
- **Out-of-Stock Substitution Engine:** Implemented a semantic-similarity substitution model to suggest alternative products when SKUs were unavailable, enhancing customer retention and reducing lost sales.
- **Incident Response & Data Recovery:** Orchestrated rapid data recovery during a cybersecurity incident, triaging and restoring 6,000 critical files across 800 billion records in two weeks alongside a colleague.
- **Technical Leadership & Mentorship:** Coached data scientists, ML engineers, and software teams on best practices in ML lifecycle (data engineering, model training, CI/CD, monitoring), and communicated complex analytics to executives.
- **Advanced Personalization & Re-Ranking Service:** Developed a dynamic personalization service that ingests product lists and re-ranks them based on user preferences across three algorithms, driving a 43% lift in add-to-cart conversions.
- **Team Culture & Agile Practices:** Promoted a positive, collaborative team culture; facilitated DS code reviews, sprint planning, and architecture design sessions; championed Agile methodologies and continuous improvement.

Senior Data Scientist Peapod Digital Labs, USA Jan 2022 – Dec 2023

- **Semantic Search with T5 Bi-Encoder:** Developed a semantic search solution for grocery queries using a T5 Text-to-Text Transfer Transformer in a bi-encoder setup, fine-tuned on domain data to boost retrieval relevance.
- **Model Distillation & Latency Optimization:** Fine-tuned and distilled the T5 model to reduce inference latency, then implemented a FAISS index to accelerate vector retrieval times for real-time analytics.
- **Deployment & API Microservice:** Developed a Dockerized Seldon microservice for semantic search inference—supporting real-time streaming deployments and batch processing by retrieving candidates from MongoDB—ensuring scalable,

low-latency production serving.

- **Cross-Sell Two-Tower Architecture**: Spearheaded design and implementation of a two-tower cross-sell model using dual embedding networks, improving product pairing precision and driving incremental revenue.
- **Strategic Roadmap & Stakeholder Alignment:** Authored the Data Science quarterly roadmap and coordinated with cross-functional product leads (Merchandising, UX, Engineering) to operationalize DS solutions.
- **Shoppable Recipe Mapping:** Developed a recipe-to-product mapping algorithm, enriching the shoppable recipe experience and increasing user engagement.
- **Trending & Popularity Recommendations:** Engineered trending and popular product recommendation algorithms that lifted revenue by 5.75%.
- **ML Demonstration Portal**: Built an interactive portal showcasing live ML algorithm outputs, enabling stakeholders to validate models and drive adoption.
- **Seasonality Detection Collaboration:** Led a joint project with Boston University, mentoring a team of four to create a transaction-history and cultural-trend-based seasonal product identification algorithm.

Data Scientist Peapod Digital Labs, USA Jan 2022 – Dec 2022

- Developed an algorithm called Predict my list for recommending the products for the users based on the history to allow faster add to cart which resulted in an increase in conversion revenue by 1.9 % by increasing conversion revenue.
- Created an algorithm for serving personalized weekly offers and coupons to different customers based on customer preferences and customer buying patterns.
- Built voice navigation system to reduce the time taken by the user to find the right product and increase the add to cart and navigate through the website and native application with voice
- Designed and built online monitoring system for measuring different KPIs (key performance indicators) to understand the performance of the ML algorithms and monitor the performance of ML Models.
- Developed a Power BI dashboard for Metrics and KPIs for reporting to the leadership of the Org
- Oversaw a team of two for above-mentioned projects and mentored two interns in developing Machine learning solutions

Data Scientist (Co-op) Retail Business Services (Ahold Delhaize), USA Jan 2021 –Sep 2021

- Designed a deep-learning classifier for 70 classes with a test accuracy of 98% using transfer learning in PyTorch on Databricks with Microsoft Vision model with fine tuning the model with less than 20 images per class
- Created a method for using a state-of-the-art Microsoft vision model for extracting features from the image and employing transfer learning using that for classification Algorithm and other models like InceptionNet, XceptionNet, and MobileNet
- Built CI/CD pipeline in Azure DevOps for training model and inference scoring using ML Flow on Databricks cluster
- Packaged and containerized models with Seldon to provide microservices with REST API and gRPC endpoint
- Functional testing of the images and deploying them as pods onto Kubernetes Cluster to provide microservice API
- Formulated tracking and logging process in ML flow for Tracking experiments to record and compare parameters and results with and used python logger for drift management and monitoring the pipeline
- Built CI/CD pipeline for embedding generation with Pre-trained Bert on Azure DevOps on Databricks cluster for search queries and products for testing developing solutions for search Algorithm
- Implemented Easy OCR and benchmarked its performance with Azure OCR and other models

Machine Learning Engineer Squark, Burlington, MA, USA (start-up) May 2020 – Jan 2021

- Designed the framework for predictive modeling with H2O, which was data agnostic and with metrics for evaluation
- Productionized the code and deployed on AWS cluster to generate predictions on the client datasets
- Programmed a solution to use the H20 data frame for Shapely value generation with a custom wrapper class
- Implemented Explainable AI technique for model using SHAP for throwing light on black box models by multiprocessing
- Created a method for SHAP generation and evaluation which brought down the time from 2 hours to 15 minutes

Machine Learning Engineer Rove Labs, Bengaluru, India (start-up) Jun 2017 – Jul 2019

- Constructed ML models for data collected from the IOT (Internet of Things) devices and sensors
- Developing a no code analytic platform too to provide predictive analytical solution
- Designed different machine learning models as proof of Concept for different client data for predictive modeling

Data Science Skills

- Generative AI (GenAI): LLMs, RAG (Retrieval-Augmented Generation), NLP, Prompt Engineering, Agentic Frameworks, Vector Databases, Fine-tuning LLMs.
- Agentic Frameworks: Multi-agent collaboration using CrewAI, LangGraph, LangChain, Ollama, LlamaIndex.
- Large Language Models (LLMs): OpenAI GPTs, Azure OpenAI, Llama 2 & 3, Mistral, Gemini, Claude, Phi-3, GPT-4o mini.
- Recommendation Systems: Neural Collaborative Filtering (NCF), Cross-sell Two-Tower models, Advanced Re-ranking systems, Personalized Coupons/Offer Recommendations.
- Graph Machine Learning: Knowledge Graphs, GraphFrames in PySpark, Similarity-Based Product Linking, Graph-based Cross-sell.
- Search Technologies: Semantic Search (T5 Bi-Encoder), Hybrid Search with Personalization, FAISS, ElasticSearch integration.
- Azure AI/ML Services: Azure Machine Learning Studio, SDK, CLI, Azure AI Services, Azure OpenAI, Databricks (Unity Catalog, Apps).
- Deep Learning & Computer Vision: PyTorch, TensorFlow, Keras, CNNs (YOLO, OpenCV object detection), Vision Transformers.
- Artificial Neural Networks (ANN): Multilayer Perceptron (MLP), Deep Learning for Recommendations & Ranking.
- Classical Machine Learning: Decision Trees, SVMs, SVR, Random Forests, XGBoost, LightGBM, Ensemble Methods.
- Programming & Tools: Python, PySpark, Git, Databricks, Distributed Training (multi-GPU), MLflow for MLops.
- EDA, Data Visualization & Dashboards: Pandas, NumPy, Scikit-learn, Power BI, Tableau, Matplotlib, Seaborn.
- MLOps & Deployment: Model tracking, versioning, packaging, registration, deployment with MLflow, Seldon Core, Docker, Azure ML pipelines.
- UI Frontend Development: Streamlit (interactive ML apps.
- Impact Measurement & KPI Definition: Metric frameworks for model impact assessment across models.

PROJECTS

Gemini Pro: Generating Embeddings and Chat Applications with Python Blog **Jan 2025**

- Blog explaining how to use Gemini Pro, to generate embeddings for text and create conversational AI applications effectively.
- Provided detailed, step-by-step instructions for developers to leverage Gemini Pro for text understanding, embedding generation, and conversational AI, making advanced AI techniques more accessible.

Open-Source Contribution - Microsoft Vision classification (PyTorch, Python) Blog **Mar 2021**

- Created a method with PyTorch for transfer learning and feature extraction from the Microsoft Vision model
- Contributed this method to Microsoft GitHub as an open-source method

EDUCATION

Stanford University - Machine Learning with Graphs Sep 2024 – Dec 2024

Harvard Business School – Management Essentials May 2024 - Sep 2024

Northeastern University, Boston, MA Jan 2019 - Dec 2021
Master of Science in Computer Software Engineering GPA 3.8